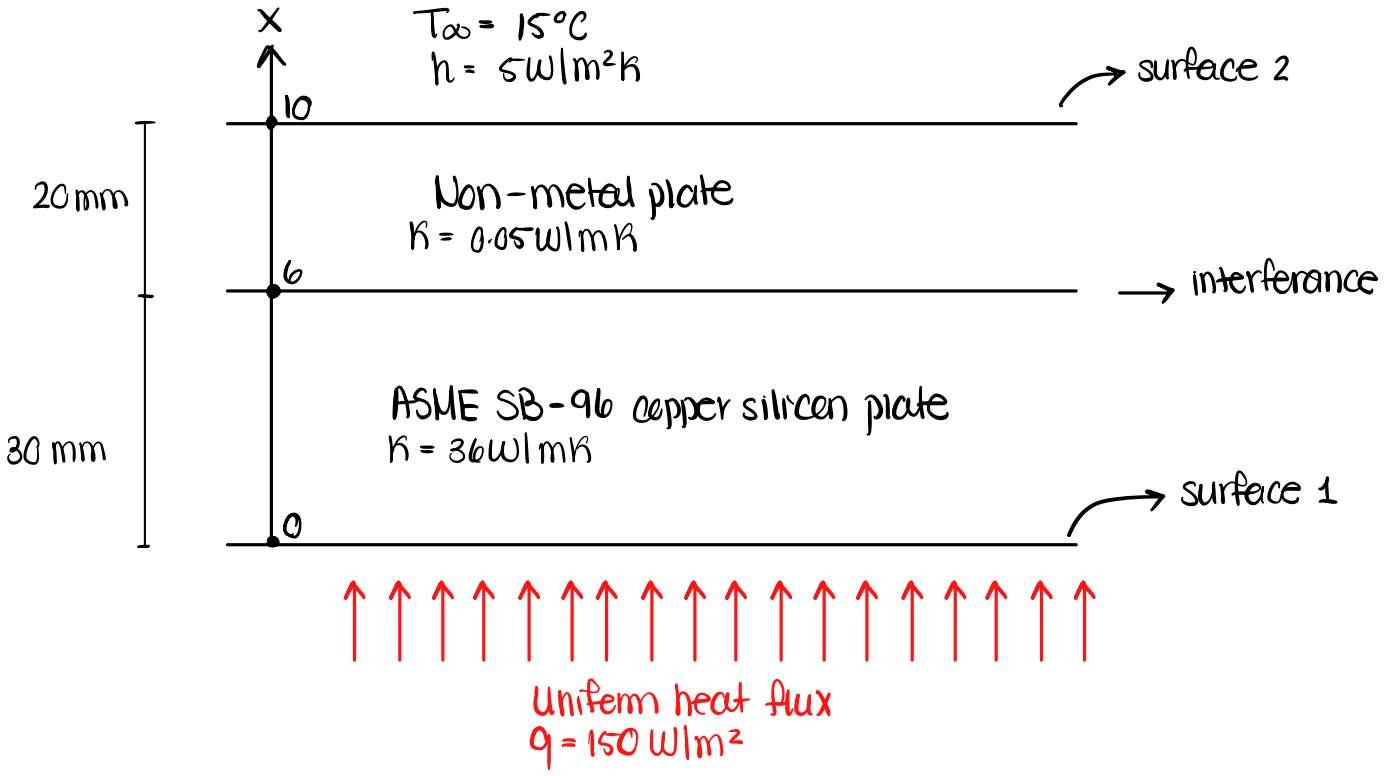
Proyecto Pt. A Heat Transfer
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Ejercicio 5-34



Boundary Conditions:

#1 $q_{cond\ metal} = q_{flux}$ @ $x=0$
#2 $q_{cond\ non\ metal} = q_{cond\ metal}$ @ $x=6$
#3 $q_{cond\ non\ metal} = q_{conv}$ @ $x=10$

#1 $q_{flux} = q_{cond\ metal}$

$q_{flux} = -k_1 \frac{\Delta T_{metal}}{\Delta X_{metal}}$

$\Delta T_{metal} = -\frac{q_{flux}\,\Delta X_{metal}}{k_1}$

#2 $q_{cond\ metal} = q_{cond\ non\ metal}$

$q_{flux} = -k_2 \frac{\Delta T_{non\ metal}}{\Delta X_{non\ metal}}$

$\Delta T_{non\ metal} = -\frac{q_{flux}\,\Delta X_{non\ metal}}{k_2}$

#3 $q_{cond\ non\ metal} = q_{conv}$

$q_{flux} = -h_{air}(T_\infty - T_{10})$

$q_{flux} = -h_{air}\,\Delta T$

$\Delta T = -\frac{q_{flux}}{h_{air}}$

- En condiciones steady-state, q (heat flux) es igual en todos los puntos
  ↳ la cantidad de energía se conserva
- Lo que cambia es el material y la gradiente de temperatura
- Cuando utilizas el método de integración, tu consideras el ΔT y ΔX total, pero en finite method consideras pedazo por pedazo

Hallando la temperatura en los diferentes nodos: $5mm\left(\frac{10^{-3}m}{1mm}\right) = 0.005m$

$$T_{10} = T_\infty - \frac{q_{flux}}{h_{air}} = 288.15K + \frac{150\,W/m^2}{5\,W/m^2\cdot K} = 318.15K = 45°C$$

$$T_9 = T_{10} + \frac{q_{flux}\,\Delta x_{non\ metal}}{k_2} = 318.15K + \frac{(150\,W/m^2)(0.005m)}{0.05\,W/mK} = 333.15K = 60°C$$

$$T_8 = T_9 + \frac{q_{flux}\,\Delta x_{non\ metal}}{k_2} = 333.15K + \frac{(150\,W/m^2)(0.005m)}{0.05\,W/mK} = 348K.15 = 75°C$$

$$T_7 = T_8 + \frac{q_{flux}\,\Delta x_{non\ metal}}{k_2} = 348.15K + \frac{(150\,W/m^2)(0.005m)}{0.05\,W/mK} = 363.15K = 90°C$$

$$T_6 = T_7 + \frac{q_{flux}\,\Delta x_{non\ metal}}{k_2} = 363.15K + \frac{(150\,W/m^2)(0.005m)}{0.05\,W/mK} = 378.15K = 105°C$$

$$T_5 = T_6 + \frac{q_{flux}\,\Delta x_{metal}}{k_1} = 378.15K + \frac{(150\,W/m^2)(0.005m)}{36\,W/mK} = 378.1708K = 105.0208°C$$

$$T_4 = T_5 + \frac{q_{flux}\,\Delta x_{metal}}{k_1} = 378.1708\,K + \frac{(150\,W/m^2)(0.005m)}{36\,W/mK} = 378.1916K = 105.0417°C$$

$$T_3 = T_4 + \frac{q_{flux}\,\Delta x_{metal}}{k_1} = 378.1916K + \frac{(150\,W/m^2)(0.005m)}{36\,W/mK} = 378.2124K = 105.0625°C$$

$$T_2 = T_3 + \frac{q_{flux}\,\Delta x_{metal}}{k_1} = 378.2124K + \frac{(150\,W/m^2)(0.005m)}{36\,W/mK} = 378.2332K = 105.0833°C$$

$$T_1 = T_2 + \frac{q_{flux}\,\Delta x_{metal}}{k_1} = 378.2332K + \frac{(150\,W/m^2)(0.005m)}{36\,W/mK} = 378.2540K = 105.1042°C$$

$$T_0 = T_1 + \frac{q_{flux}\,\Delta x_{metal}}{k_1} = 378.2540K + \frac{(150\,W/m^2)(0.005m)}{36\,W/mK} = 378.2748K = 105.1250°C$$

b. Under these conditions, it will not be in compliance as $105.1250°C > 93°C$

c. h_{conv} needed to maintain $T_0 < 93°C$

$$\Delta T_{total} = q_{flux}\left(\frac{1}{h} + \frac{L_1}{k_1} + \frac{L_2}{k_2}\right)$$

$$\frac{1}{h} = \frac{\Delta T_{total}}{q_{flux}} - \frac{L_1}{k_1} - \frac{L_2}{k_2}$$

$$\frac{1}{h} = \frac{366.15K - 288.15K}{150\,W/m^2} - \frac{0.030m}{36\,W/mK} - \frac{0.020m}{0.05\,W/mK}$$

$$\frac{1}{h} = 0.11917\,\frac{m^2\cdot K}{W}$$

- Changing the materials, and overall the sample is not always feasible. Therefore, you should change the experimental conditions.

$h = 8.39161 \frac{W}{m^2 \cdot K}$ → increasing h will cool down the system
(forced convection ~ increasing fluid flow above the material to allow more heat to escape

Manipulating physical parameters for faster cooling:

① Forced convection: increasing h

$$\Delta T_{total} = q_{flux}\left(\frac{1}{h} + \frac{L_1}{K_1} + \frac{L_2}{K_2}\right)$$
$$T_0 = T_\infty - q_{flux}\left(\frac{1}{h} + \frac{L_1}{K_1} + \frac{L_2}{K_2}\right)$$

taking $h_{forced} = 5h_{given} = 25\ W/m^2K$, and $T_\infty = 15°C = 288K$

$$T_0 = 288.15K + 150\frac{W}{m^2}\left(\frac{1}{25\ W/m^2K} + \frac{0.030m}{36\ W/mK} + \frac{0.020m}{0.05\ W/mK}\right)$$
$$T_0 = 354.275K$$
$$= 81.1250 \quad < 93°C$$

↳ Increasing h by forcing convection allows more heat to escape the system and cool faster.
The system will be in compliance with the regulations.
(you are holding T_∞ constant, therefore Q will change)

② Changing material to a metal: increasing thermal conductivity

$$\Delta T_{total} = q_{flux}\left(\frac{1}{h} + \frac{L_1}{K_1} + \frac{L_2}{K_2}\right)$$
$$T_0 = T_\infty + q_{flux}\left(\frac{1}{h} + \frac{L_1}{K_1} + \frac{L_2}{K_2}\right)$$

$K_{iron\ (transition\ metal)} = 80.2\ W/mK$, and assuming $T_\infty = 15°C = 288K$

$$T_{10} = 288K + 150\frac{W}{m^2}\left(\frac{1}{5\ W/m^2 \cdot K} + \frac{0.030m}{36\ W/m \cdot K} + \frac{0.020m}{80.2\ W/m \cdot K}\right)$$
$$T_{10} = 318.312K$$
$$= 45.162\ °C$$

↳ Changing the non metal for a metal will allow more heat to flow, allowing the system to cool down faster.
This will allow the system to be in compliance with the regulations.
(you are holding T_∞ constant, therefore Q will change)